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Question Paper Code : 20613

B.E. / B.TECH. DEGREE EXAMINATIONS, NOVEMBER / DECEMBER 2023

Third Semester

Computer Science and Engineering

19CS301 – DATA STRUCTURES AND ALGORITHMS

(Regulations: Mepco – R2019)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

BTL, CO

R, CO1 1. Represent the notion of the algorithm diagrammatically.

U, CO1 2. If the time complexity is given by the order of growth O(2^n), then the complexity of it is _____. Justify your answer.

A) Constant B) Quadratic C) Linear D) Exponential

U, CO2 3. Which of the following data structures is best suited for implementing a recursive algorithm? Justify your answer.

A) Array B) Linked list C) Queue D) Stack

A, CO2 4. Let a = 4, b = 3, c = 2 and d = 5. Determine the output of the following postfix expression after evaluation: abc*+d-.

A, CO3 5. Determine the minimum number of nodes in a full binary tree of depth 4. Justify your answer.

A) 9 B) 14 C) 15 D) 16

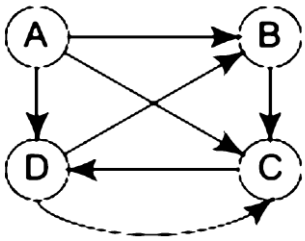
U, CO3 6. Identify any two applications of Priority Queue.

R, CO4 7. When two sets A and B are said to be disjoint sets?

R, CO4 8. If several elements are competing for the same bucket in the hash table, what is it called? Justify your answer.

A) Congestion B) Collision C) Diffusion D) Replication

A, CO5 9. Derive the adjacency matrix and adjacency list representation of the following graph:



U, CO5 10. A graph has an Euler circuit if _____. Justify your answer.

A) Every vertex has even degree

B) It is connected and has an even number of vertices

C) It is connected and has an even number of edges

D) It is connected and every vertex has even degree

PART B – (5 × 16 = 80 Marks)

A, CO1 11. a) For each of the following algorithms:

A) Compute the average of n numbers

B) Find the smallest element in a list of n numbers

C) Reverse a list of n numbers

D) Search for a given value in an array of n numbers

Analyze the above algorithms and indicate:

A) Basic Operation(s)

B) Natural size metric for its inputs

C) Whether the basic operation count can be different for inputs of the same size

D) Derive Big-Oh, Big-Omega and Big-Theta Notation (16 Marks)

OR

A, CO1 11. b) Outline the general framework for investigating recursive algorithms. Apply it to find the number of binary digits in the binary representation of a positive decimal integer and express its time complexity using Big-oh, Omega and Theta notations. (16 Marks)

U, CO2 12. a) Design an algorithm to implement the Doubly Linked List ADT for the following operations and illustrate it diagrammatically:

A) Insert at last

B) Delete the given element

C) Display the element in the list (16 Marks)

OR

U, CO2 12. b) Provide the suitable code to implement the basic operations on Queue using Linked list and array. (16 Marks)

A, CO3 13. a) The in-order and post-order traversal of a tree is given below:

In-order: DBMINEAFCJGK

Post-order: ABDEIMNCFGJK

A) Construct the corresponding binary tree

B) Determine the post-order traversal of the tree

Provide the recursive code for the tree traversal. (16 Marks)

OR

A, CO3 13. b) Perform the following operation:

A) Insert the following sequence of elements into an AVL tree, starting with an empty tree:

30, 25, 10, 5, 7, 50, 42, 19, 15, 23, 22

B) Delete 15 from the above constructed AVL tree.

Print the constructed AVL tree with the balance factor at each and every insertion and deletion of an element and provide the code for the same. (16 Marks)

A, CO4 14. a) Provide a suitable pseudocode to illustrate an efficient data structure used to solve the equivalence problem. (16 Marks)

OR

A, CO4 14. b) Given the values {2341, 4234, 2839, 430, 22, 397, 3920} a hash table of size 7 and a hash function h(x) = x mod 7, show the resulting table after inserting the values in the given order with each of the following collision strategies:

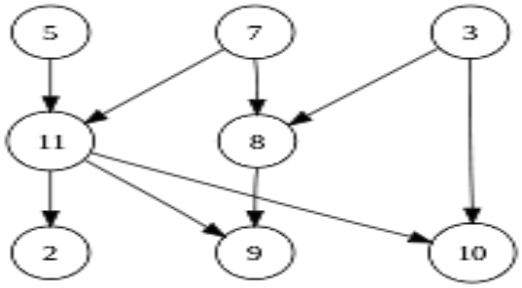
A) Separate chaining

B) Linear probing

C) Double hashing with second hash function

h1(x) = (2x – 1) mod 7. (16 Marks)

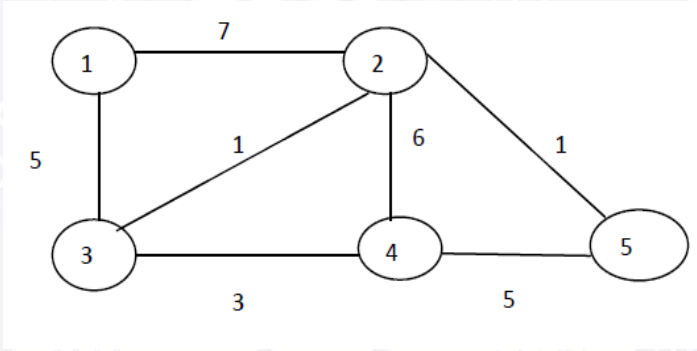
A, CO5 15. a) Write down the pseudo code to perform the topological sort. Perform topological sort on the following graph and show the result of applying this sort:



(16 Marks)

OR

A, CO5 15. b) Illustrate Prim's and Kruskal's algorithm to find the minimum spanning tree of the following graph and write the algorithms:



(16 Marks)